

Emerging Technologies in Finance: The Imperatives of Purpose, Prudence, and Policy

(Keynote Address by Shri Rohit Jain, Deputy Governor, Reserve Bank of India at the Global Fintech Fest, 2026 in Mumbai on September 9, 2026)

Distinguished guests, colleagues from across the fintech ecosystem, representatives of regulatory and international bodies, ladies and gentlemen. It gives me great pleasure to be here at the Global Fintech Festival.

2. This Festival has, over the years, become a fertile ground for ideas at the intersection of finance and technology. It brings together those who build, those who use, and those who shape the frameworks in which innovation takes place. That makes it an especially appropriate forum to reflect not only on where technology is taking finance, but also on the choices we make along the way.

3. Every **major technological wave has expanded the range of what human beings can do**. Steam power helped spark the Industrial Revolution by replacing much manual and animal labour with machines. Electricity brought instant lighting, cooling and a host of conveniences that transformed homes, workplaces, and factories. Computers greatly expanded our ability to calculate, process and store information. The internet then connected people, businesses, and markets across distances almost instantaneously. Each of these technologies changed not only how we lived and worked, but also how economic activity was organised.

4. The technological wave before us today takes this progression a step further. Artificial intelligence is not only helping us process information faster; it is increasingly helping us interpret information, identify patterns, make predictions and support decisions. In that sense, **AI is beginning to augment something especially consequential for finance: human judgment**.

5. Artificial intelligence, however, is only one part of a much wider technological transformation. Tokenisation is creating new possibilities for how financial assets are represented and settled. Distributed technologies are reshaping elements of market infrastructure. Quantum computing holds promise for solving complex problems, while also raising new questions about digital security and cryptographic resilience.

6. Alongside these emerging technologies, digital public infrastructure is changing the foundations on which financial services can be delivered. Digital identity and interoperable payments have enabled financial services to reach customers in ways that were difficult to imagine even a decade ago. Technology is therefore changing both the services we offer and the infrastructure on which finance operates.

7. Taken together, these developments present enormous possibilities. They can lower costs, widen access, strengthen risk management, and make finance more responsive. **They also raise important questions about resilience, accountability, and the choices we make as these technologies scale.** This brings me to the three imperatives I would like to discuss today: Purpose, Prudence, and Policy.

8. I would like to approach these developments through three questions.

- (i) First, **what purpose should technology ultimately serve** in finance?
- (ii) Second, how do we ensure that **greater speed, scale and autonomy are matched by adequate safeguards?**
- (iii) Third, how should **policy evolve so that innovation is encouraged without compromising resilience, fairness and trust?**

9. These questions correspond to the three themes of my remarks: Purpose, Prudence and Policy. They are closely connected.

- (i) Purpose tells us what is worth pursuing.
- (ii) Prudence asks what we must protect as innovation scales.
- (iii) Policy provides the framework within which both can advance together.

Purpose: Technology as a means, not an end

10. There is an old story about a young Boy Scout determined to complete his good deed for the day. He spots an elderly lady standing hesitantly at a busy street corner and immediately decides that she needs help crossing the road. Despite her protests, he takes her firmly by the arm and, with considerable effort, gets her safely to the other side. Quite pleased with himself, he asks whether she is all right. The lady looks at him and says, "I was all right until you brought me here. I had crossed to the other side with great difficulty, and you have just brought me back to where I started!"

11. There is a lesson here for innovation as well. A solution, however well-intentioned or technologically impressive, **has little value if it does not address the problem that actually needs solving**. Technology should therefore remain a means to an end. The starting point has to be the purpose it is intended to serve.

12. From ancient Mesopotamia nearly 4,000 years ago to the financial institutions of today, the forms of finance have changed beyond recognition. The **essential purposes, however, have remained remarkably familiar**: helping people save, make payments, borrow and invest, and protect themselves against uncertainty and risk.

13. If the purpose has endured, then that purpose should also be our reference point when evaluating new technology. We all know that emerging technologies can make financial services faster, cheaper and more accessible. **The more important question is whether they solve a genuine financial problem better, who benefits from that improvement, and what new costs or risks may arise in the process.**

14. These questions need to be considered at design stage itself. A technology may lower costs and remove friction, yet those gains will not necessarily reach everyone. Whether innovation broadens participation depends on choices such as interoperability, common standards, accessibility and the economics of serving smaller customers. Wider participation, therefore, often has to be designed for rather than assumed.

15. UPI offers a clear illustration. It was built around interoperability, allowing customers to make payments across participating banks and applications rather than remain within closed networks. That design choice widened convenience for users and created a common payment infrastructure on which banks, fintechs and other service providers could innovate.

16. The value of such design is ultimately seen in the experience of the user. A small merchant can receive payment instantly, a customer can transact without worrying about which bank or application the other person uses, and everyday payments become simpler and more convenient. Technology becomes meaningful when these improvements translate into wider participation and greater economic opportunity.

17. Purpose, therefore, gives us the first discipline for approaching emerging technology: begin with the need, not the novelty. **Yet even a technology that solves the right problem can create new vulnerabilities when it operates at greater speed, reaches larger scale or becomes more interconnected.** That brings me to the second imperative: Prudence.

Prudence: When scale changes the nature of risk

18. I see **three key concerns** as emerging technologies become more deeply embedded in finance: **speed, concentration and opacity.** None of these risks is entirely new, but technology can amplify them and allow their effects to travel through the financial system in ways that are faster, wider and sometimes harder to detect.

19. The first concern is **speed.** Automated systems can analyse information and initiate actions far faster than human beings can respond. At machine speed, resilience cannot depend only on preventing every error. Institutions must also be able to detect problems early, contain their effects and intervene before a small mistake becomes a much larger one.

20. The second concern is **concentration.** Financial institutions may increasingly depend on a relatively small number of cloud providers, technology vendors and model providers, often using overlapping datasets and similar technological infrastructure. The concern is therefore not simply the failure of one institution, but the possibility that a common dependency could transmit disruption or error across many institutions at the same time.

21. The third concern is **opacity.** Advanced models can identify relationships and arrive at decisions in ways that may be difficult to explain. Greater sophistication, however, cannot mean weaker accountability. An institution may outsource the computation, but it cannot outsource the consequence. A customer affected by an important financial decision deserves something more meaningful than being told that “the model said so”.

22. These concerns arise in a financial system whose underlying risks remain familiar. Borrowers can still default, liquidity can still disappear, leverage can still magnify losses, and operational failures can still disrupt financial services. **Technology does**

not make these risks vanish. Instead, what it can change, sometimes significantly, is their speed, scale and transmission.

23. Prudence must also extend to risks that may not yet be immediate. Quantum computing offers a useful example. It holds significant promise, but it also raises questions about the resilience of current cryptographic systems. Preparing in advance reflects a broader principle: we should not wait for a future vulnerability to become a present crisis before responding.

24. Prudence, therefore, is not about resisting innovation. It is about ensuring that innovation remains resilient as it scales. Yet this creates a difficult question for policymakers. Technology may evolve faster than our ability to foresee all its consequences. **When should policy intervene, and how can it do so without closing off useful innovation?**

Policy: Finding the right balance

25. This is not an easy balance to strike. **Regulate too early, and we risk writing detailed rules for a technology we do not yet fully understand,** or for an architecture that may change before the rules take effect. **Regulate too late, and the technology may already be deeply embedded** before its risks are fully understood and addressed. There is no perfect point between these two outcomes.

26. Policy therefore needs both **conviction and humility**. **Conviction** is required about the outcomes that matter, including fairness, accountability, resilience, customer protection and financial stability. **Humility** is equally necessary about our ability to predict how a new technology will evolve or where its most valuable applications will eventually emerge.

27. I see three elements as important in navigating this balance.

28. The first is to **remain clear about outcomes and accountability** rather than attempt to prescribe every technological choice. The obligation to treat customers fairly does not change because an algorithm influences the decision. Similarly, responsibility for managing risk does not disappear because a model or technology is supplied by a third party.

29. The second element is **proportionality**. Different uses of the same technology can present very different risks. A tool used to summarise an internal document cannot be treated in the same way as a system that autonomously approves credit or executes financial transactions. The greater the consequence of the use case, the stronger the expectations should be around governance, validation, oversight and intervention.

30. The third element is to **create space to experiment and learn within appropriate safeguards**. Emerging technologies are often understood better through carefully controlled use than through speculation alone. Regulatory sandboxes can play an important role here by allowing genuinely new applications to be tested within defined boundaries before they are deployed more widely.

31. **Policy also has to remain informed by what is happening on the ground.** In a rapidly changing environment, regulators cannot understand emerging technologies through returns and supervisory observations alone. Regular engagement with financial institutions, fintechs and technology providers helps identify new use cases and emerging concerns early, while also giving industry greater clarity about regulatory expectations.

32. There is a final implication for regulators themselves. As financial institutions become more technologically capable, **supervisory capability must evolve alongside** them. Initiatives such as DAKSH and PRAVAAH illustrate how technology can improve supervisory and regulatory processes. The Digital Payments Intelligence Platform (DPIP) extends this approach by recognising that payment fraud increasingly operates across institutional boundaries and therefore calls for network-level intelligence and near-real-time information sharing.

33. The objective of policy, therefore, is to create the conditions in which useful innovation can develop responsibly. This requires clear guardrails where the risks are understood, room for experimentation where they are still emerging, and the ability to adapt the framework as technology and its uses evolve. Good policy should give innovation room to grow, while ensuring that accountability and resilience grow with it.

Conclusion

34. As I conclude, let me leave you with one final thought.

35. Amid all our discussions about algorithms, tokens, platforms, cloud infrastructure and quantum computing, **let us not forget that there is always someone at the other end of the technology**. There is a saver entrusting an institution with hard-earned money, a borrower seeking an opportunity, a merchant awaiting a payment, or a family depending on the financial system when it matters most. That is ultimately where our responsibility lies.

36. Most customers will never know which model made a recommendation, which cloud hosted it or which technology enabled a transaction. They will, however, experience the outcome. Their **confidence in technology** will ultimately depend not on how sophisticated it is, but on **whether it works for them fairly, reliably and safely**.

37. Purpose, Prudence and Policy therefore have to move together. **Purpose without prudence can become recklessness. Prudence without purpose can become stagnation. Policy is what binds the two together at scale**. The objective should not merely be to make finance faster or smarter, but to ensure that technological progress makes finance more useful, resilient and responsive to those it serves.

38. Every technological wave we spoke about at the beginning expanded the range of what human beings could do. The opportunity before us is to ensure that this wave expands not only what finance can do, but also what finance can do better for those it serves. If we can achieve that, emerging technology will have served not merely innovation, but the larger purpose of finance itself.

39. With this, I thank the organisers of the Global Fintech Festival for the opportunity to share these thoughts, and I wish all of you engaging and productive discussions over the course of the Festival. Thank you.
